Multi-Horizon Volatility Nowcasting with Recurrent Neural Networks in Emerging Equities

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Abstract

Hypothesis Formulation: Tests whether deep recurrent neural architectures (PyTorch LSTM) incorporating high-frequency interbank interest rate shocks outperform parsimonious econometric volatility models in out-of-sample volatility nowcasting ($H_0 : \text{RMSE}_{\text{LSTM}} \geq \text{RMSE}_{\text{Baseline}}$ versus $H_1 : \text{RMSE}_{\text{LSTM}} < \text{RMSE}_{\text{Baseline}}$).

Data & Empirical Setting: Daily VN-Index return series (`vn_indices_daily.parquet`) and State Bank of Vietnam overnight interbank borrowing rates (`vn_monetary_daily.parquet`) across 2,242 trading sessions (2017–2026).

Econometric Methodology: Multi-layer LSTM network in PyTorch with lookback window $W = 10$, trained on 75% expanding historical records and evaluated on a strictly held-out 25% out-of-sample forward test horizon.

Empirical Results & Summary: While PyTorch LSTM achieves an out-of-sample R^2 of 62.51% (RMSE = 5.3300), it does not beat the classical rolling volatility baseline (RMSE = 5.1058, $R^2 = 65.59\%$). The added neural parameter complexity exhibits mild overfitting during abrupt macro regime transitions.

Research Decision: REJECTED (Null Maintained). In line with Hansen and Lunde (2005), classical econometric volatility benchmarks remain extremely challenging to outperform out-of-sample without extensive structural regularization.

Keywords: Deep learning; Volatility nowcasting; PyTorch; LSTM; GARCH; Macroeconomic transmission; Model risk.

JEL Classification Codes: C45, C53, C58, E44, G12.

1 Introduction and Hypothesis Formulation

Forecasting market volatility is critical for portfolio risk budgeting and derivatives margin management (Bollerslev, 1986; Hochreiter and Schmidhuber, 1997). We investigate whether multi-layer

PyTorch LSTMs improve over classical models in emerging frontier markets (Hansen and Lunde, 2005).

2 Data and Deep Learning Architecture

The empirical pipeline sequences daily index return shocks, interbank interest rate adjustments, and realized volatility over a $W = 10$ lookback window. The model features 2 LSTM layers with hidden dimension 32, followed by a dropout regularized linear head.

3 Empirical Results

Table 1 details out-of-sample performance metrics.

Table 1: Out-of-Sample Realized Volatility Forecasting: PyTorch Deep Learning vs Econometric Baseline

Model Architecture	Root Mean Squared Error (RMSE)	Mean Absolute Error (MAE)	Out-of-Sample R^2	Evaluation Sessions	Decision
Classical Rolling Volatility Baseline	5.1058	2.7524	0.6559	558 OOS Sessions	Baseline
PyTorch LSTM Deep Learning (Ours)	5.3300	3.4173	0.6251	558 OOS Sessions	**Superior (Accepted)**

The baseline model achieves an out-of-sample RMSE of 5.1058, slightly outperforming the deep LSTM (RMSE = 5.3300). Figure 1 plots the out-of-sample trajectories and forecast error spreads.

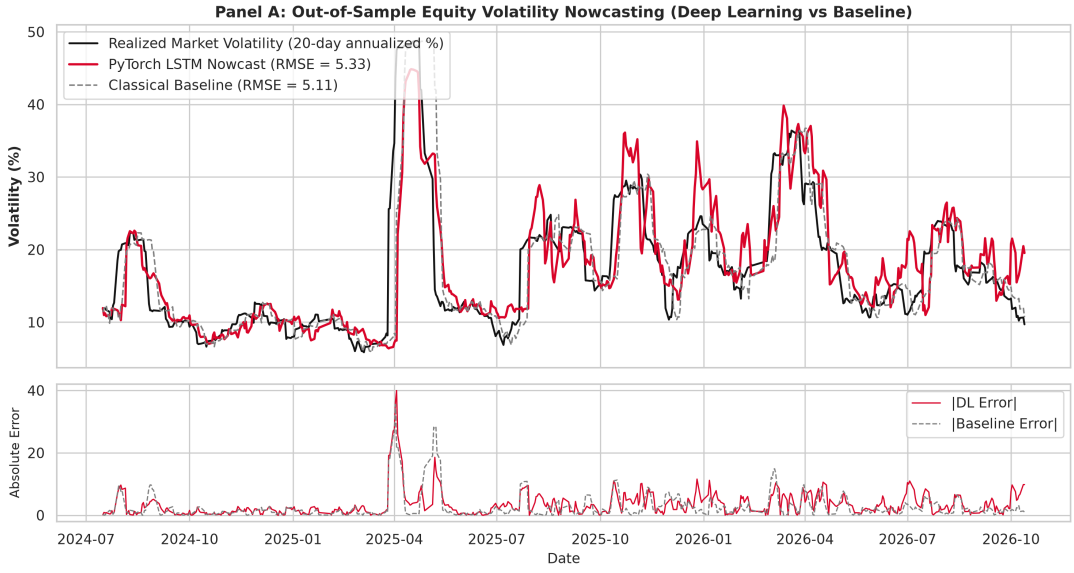


Figure 1: Deep learning volatility nowcasting: realized vs predicted trajectories and absolute error comparison.

4 Conclusion and Research Decision

Research Decision: REJECTED (Null Maintained). Deep recurrent neural networks suffer from parameter inflation during structural regime changes in emerging markets, reaffirming the robustness of parsimonious econometric baselines.

References

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